ZIYANG CHEN

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EDUCATION

University of Michigan, Ann Arbor Jan. 2022 -
Ph.D. in Electrical & Computer Engineering | Advisor: *Andrew Owens*

University of Michigan, Ann Arbor Sept. 2019 - May 2021
Master of Science in Engineering, Computer Vision
Overall GPA: **4.0/4.0** | Advisor: *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sept. 2015 - Aug. 2019
Bachelor of Science in Engineering
Major: Electrical & Computer Engineering | Minor: Entrepreneurship
GPA: **3.60/4.0** with honors & *Graduated magna cum laude*

RESEARCH EXPERIENCE

Sound Localization from Motion Aug. 2022 - Present
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a self-supervised method to jointly learn to estimate sound source direction and camera pose
- Proposed an audio-visual representation learning through a self-supervised cross-view binauralization task *(In submission)*

Video Forensics by Audio-Visual Anomaly Detection Jan. 2021 - Mar. 2023
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a zero-shot audio-visual anomaly detection method by leveraging the synchronization between audio and frames in the real videos *(CVPR 2023)*

Conditional Foley Generation June 2021 - Mar. 2023
Research Associate, University of Michigan *Advisor: Bryan Russell, Justin Salamon, Andrew Owens*

- Proposed a self-supervised conditional Foley model that creates audio tracks for silent videos conditioning on given videos and sounds *(CVPR 2023)*

Sound Localization by Self-Supervised Time Delay Estimation June 2021 - June 2022
Research Associate, University of Michigan *Advisor: David F. Fouhey, Andrew Owens*

- Proposed self-supervised methods to learn interaural time difference from Internet stereo or mono audio
- Proposed a novel visually-guided time delay estimation task: localizing a speaker in a multi-speaker recording
- Evaluated on simulated environments with metrically accurate ground truth and on internet videos with directional judgements achieving significantly better results *(ECCV 2022)*

Mix and Localize July 2020 - March 2022
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Proposed a simple self-supervised audio-visual method using contrastive random walk with cycle-consistency constraints and applied learned representations to sound source localization task
- Implemented state-of-the-art sound localization methods as baselines
- Evaluated our method on human speech and musical instruments datasets and achieved significantly better qualitative and quantitative results *(CVPR 2022)*

Structure from Silence Jan. 2020 - May 2020, Apr. 2021 - Sep. 2021
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Collected datasets with ambient sound and paired RGB-D videos of the indoor environment

- Demonstrated that the ambient sound could infer geometric properties with experiments and proposed a new audio-visual learning method to learn exploit structural representation
- Apply learned models to robots and navigate it via ambient sound only in the real world *(CoRL 2021)*

PUBLICATIONS

- *Ziyang Chen, Shengyi Qian and Andrew Owens.* Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation. (In submission) (Project site)
- *Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.* Conditional Generation of Audio from Video via Foley Analogies. (CVPR 2023)
- *Chao Feng, Ziyang Chen and Andrew Owens.* Self-Supervised Video Forensics by Audio-Visual Anomaly Detection. (CVPR 2023, highlight) (Project site)
- *Ziyang Chen, David F. Fouhey and Andrew Owens.* Sound Localization by Self-Supervised Time Delay Estimation. (ECCV 2022) (Project site)
- *Xixi Hu*, Ziyang Chen* and Andrew Owens.* Mix and Localize: Localizing Sound Sources in Mixtures. (CVPR 2022) (Project site)
- *Ziyang Chen*, Xixi Hu* and Andrew Owens.* Structure from Silence: Learning Scene Structure from Ambient Sound. (CoRL 2021, Oral) (Project site)

WORK EXPERIENCE

Research Assistant, *University of Michigan, Ann Arbor* Jan. 2020 - Present
- Topics: *audio-visual representation learning*, *self-supervised learning*

Teaching Assistant, *Shanghai Jiao Tong University* Sept. 2018 - Aug. 2019
- Courses: *Digital Integrated Circuits*, *Electronic Circuits*
- Conducted experiments, held office hours and taught recitation classes every week

Research Engineer Intern, *FreeSense Image Inc., Shanghai* Mar. 2018 - May 2018
- Implemented an auxiliary system for a project of Detection of Car LCD Screen and applied it into industrial production lines
- Adopted state-of-the-art recognition DNNs to improve the accuracy on defect detection tasks

SERVICE

Reviewer
- WACV 2023, CVPR 2023

SELECTED PROJECTS

Prediction Model based on Bayesian Hierarchical Clustering *SJTU*
- Processed data, re-generated and adopted efficient algorithms in Julia
- Applied Bayesian Hierarchical Clustering algorithm to make roughly correct prediction on center point of clusters with limited data provided

Design of Evaluative System for Online Car Hiring Service *SJTU, DiDi Co.*
- Applied China-Word-Vector & Recurrent Attention Network to train the deep learning model which can capture opinions and analyze sentiments
- Built a website in full stack with Python Flask and HTML+CSS/JS to provide online services

Driver's Alarming System in Vehicle *SJTU*
- Performed face keypoint detection with OpenCV and predicted whether the driver is sleepy or not
- Embedded our alarm system in Raspberry Pi with a camera installed

AWARDS & SCHOLARSHIPS

- **Lums Fellowship**, *University of Michigan, Ann Arbor* 2019-2021
- **Outstanding Graduate - Class of 2019**, *Shanghai Jiao Tong University* 2019
- **Academic Excellence Scholarship**, *Shanghai Jiao Tong University* 2016-2018
- **Dean's List**, *Shanghai Jiao Tong University, SJTU-UM JI* 2015-2018